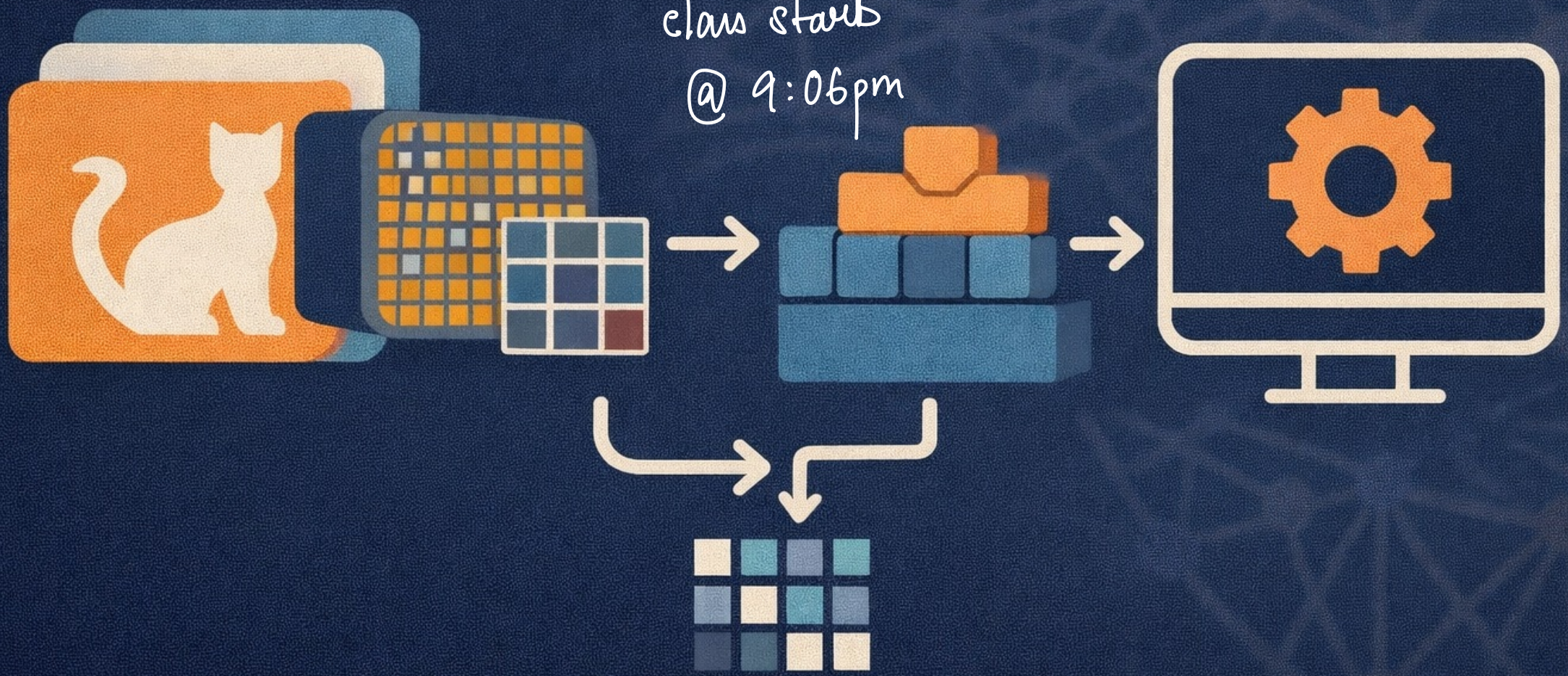
CONVOLUTIONAL NEURAL NETWORK



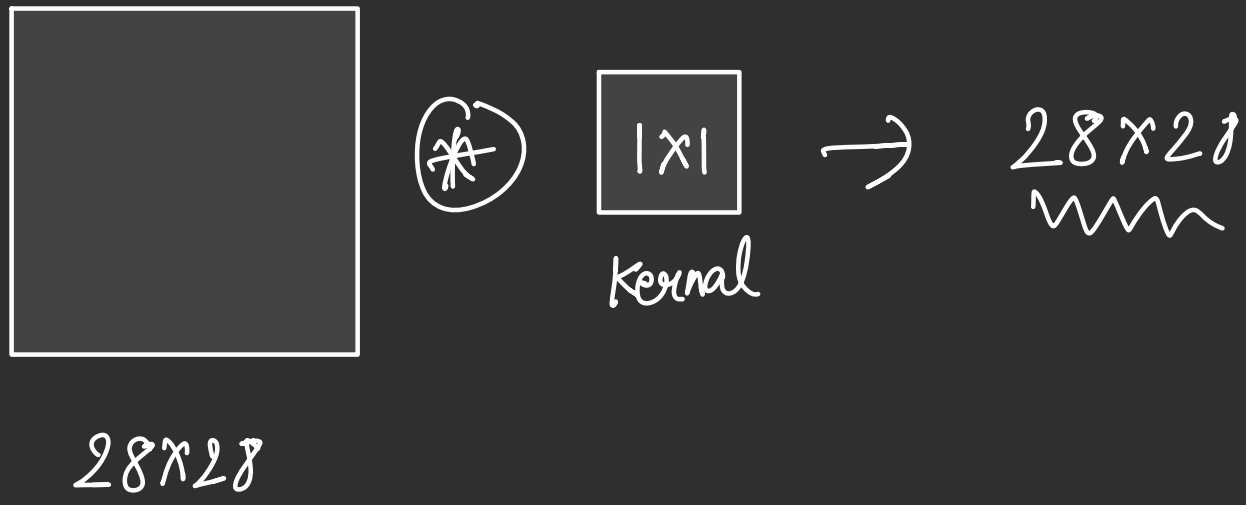
Lecture : Image Similarity using CNN

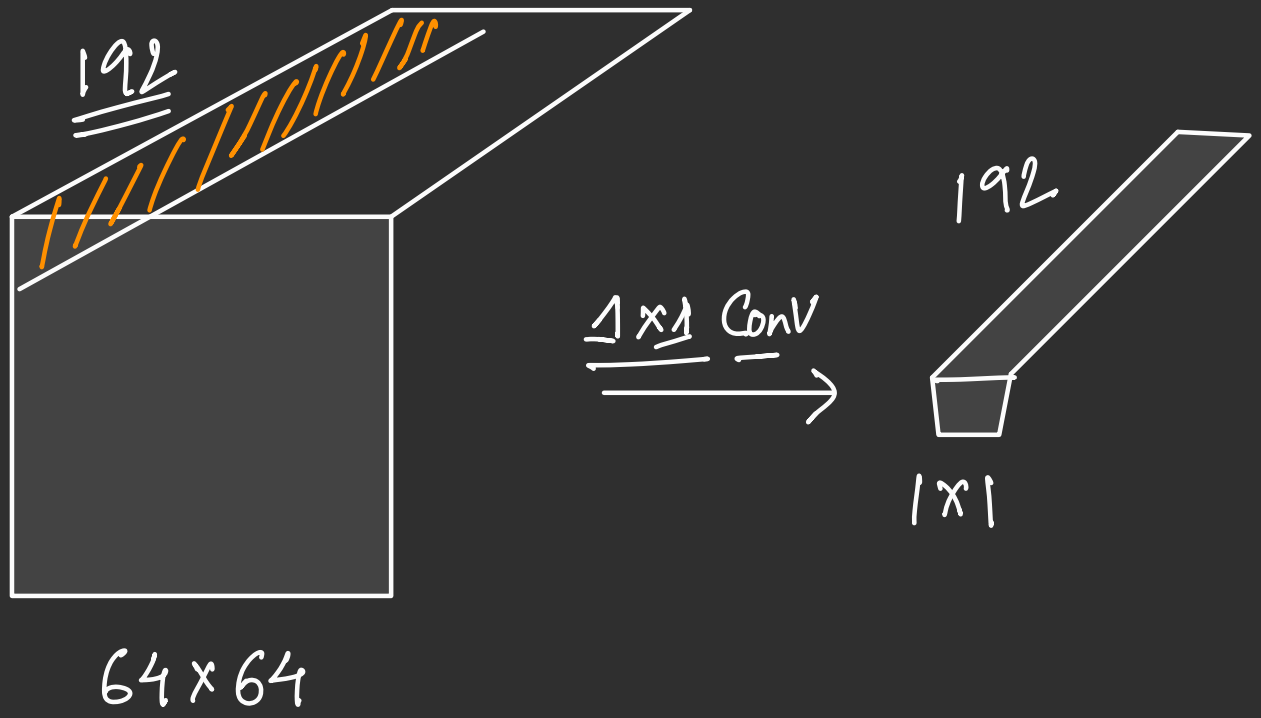
#Agenda

1) Solving a image similarity problem
2) InceptionNet
3) ResNet

Inception Net

1. Conv → 5×5, 3×3, 1×1
2. pooling

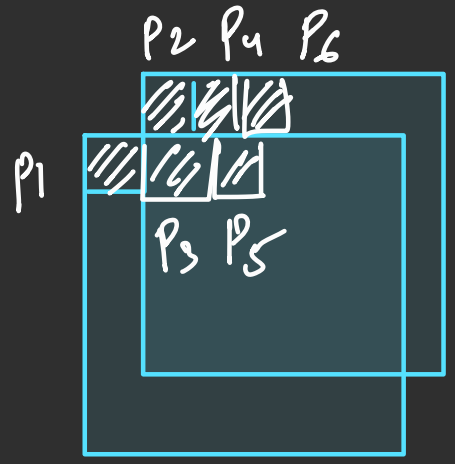




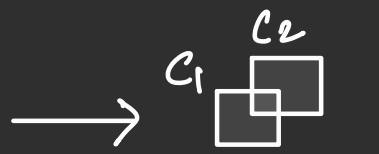
64 x 64 x 192

"192 no of filters"

2 filters



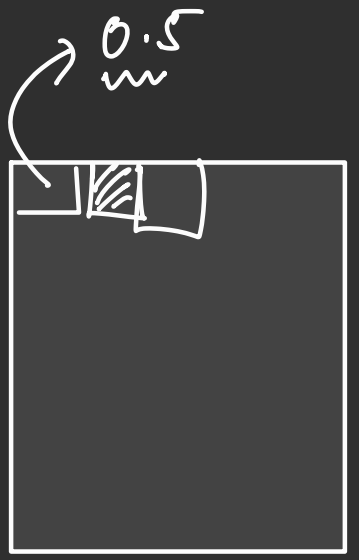
64 x 64 x 2



1 x 1 x 2

$P_1 \times C_1 + P_2 \times C_2$





64 x 64

$P_1 = 0.5 \quad C_1 = 1$

$P_2 = 0.8 \quad C_2 = 0$

$\Rightarrow (0.5 \times 1) + (0.8 \times 0)$

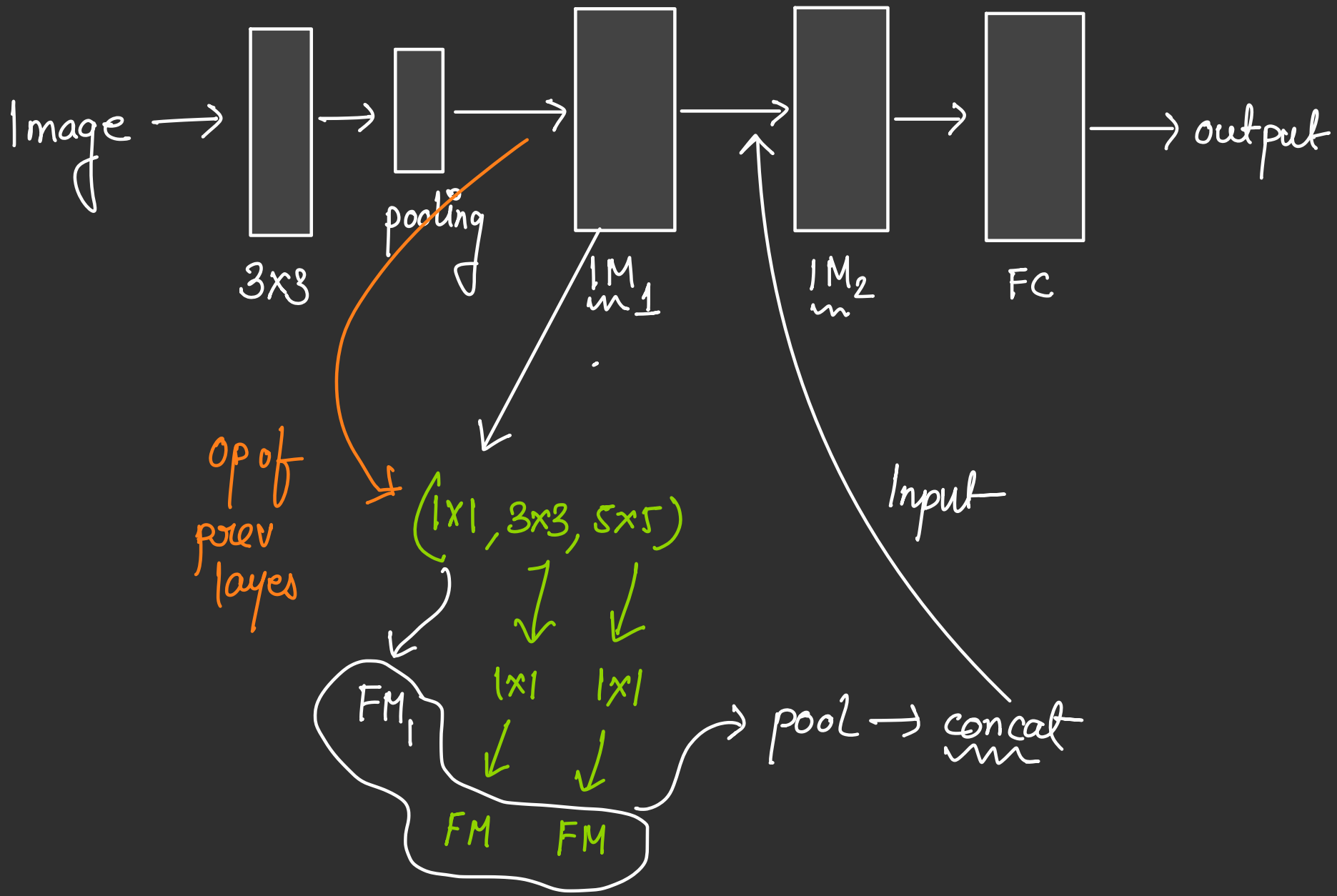
$\Rightarrow \underline{0.5}$

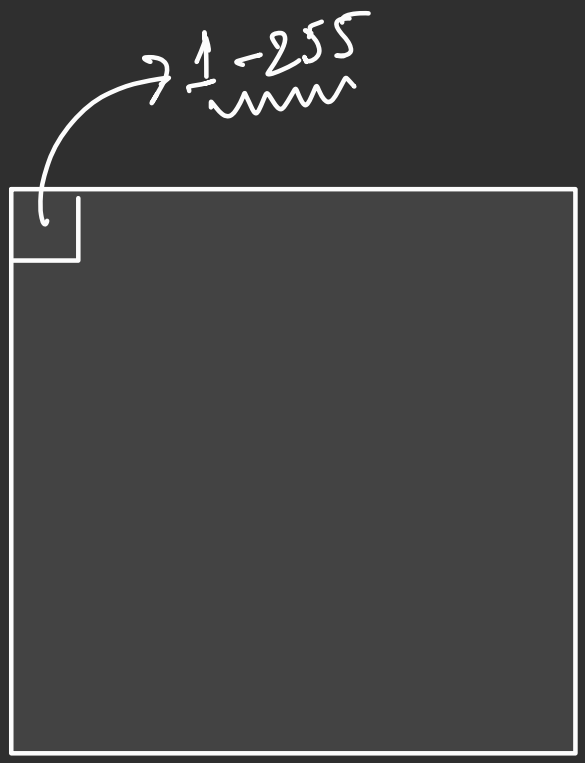
$\Rightarrow P_3 \times C_1 + P_4 \times C_2 \Rightarrow$

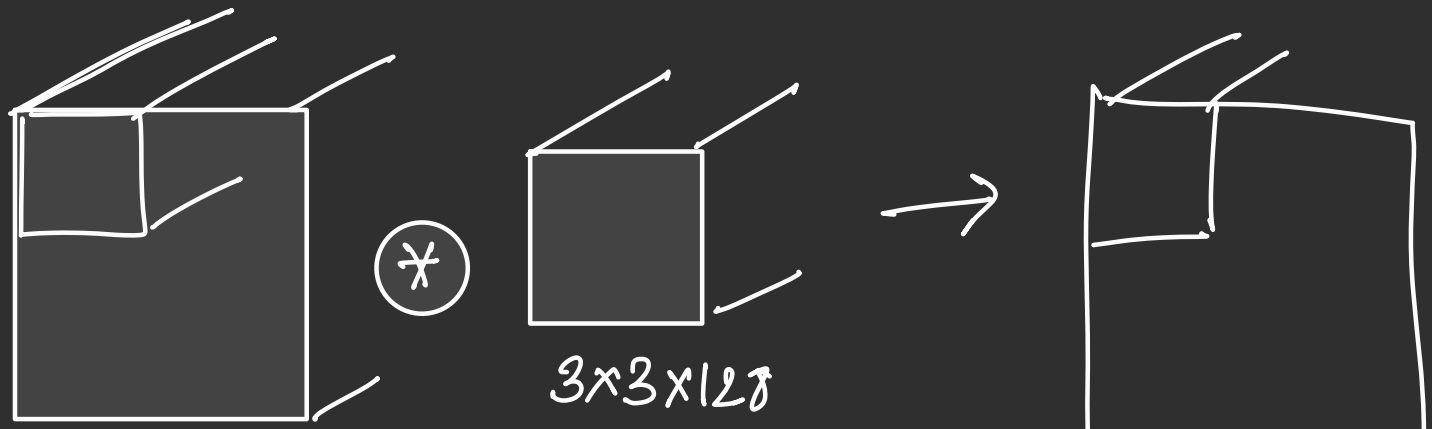
1 × 1

- Understand local features
- Reduce the size of feature map

Inception Net




1-255
64x64
size



↳ depth / No of filter

|X|

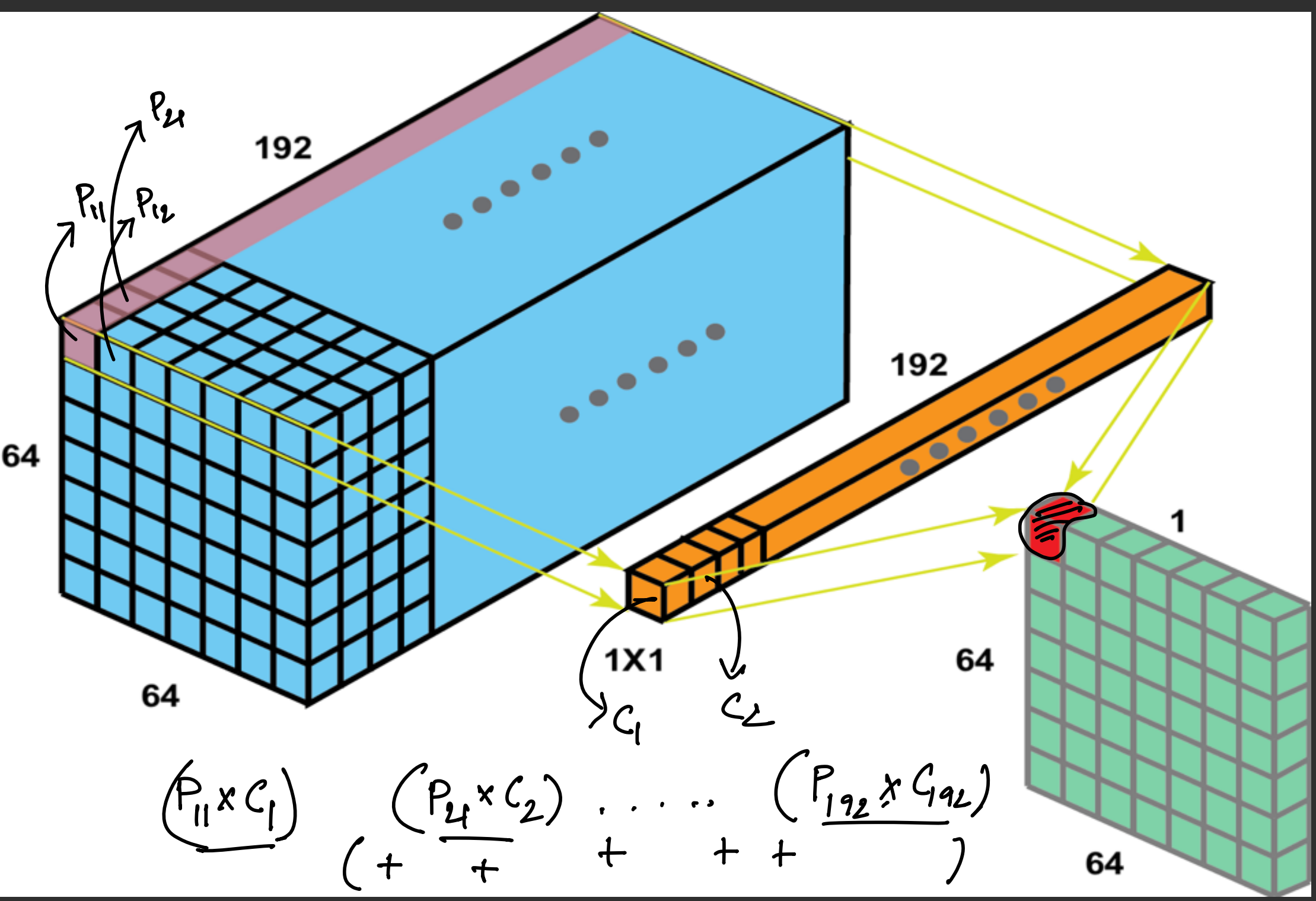
$64 \times 64 \times 256 \quad \oplus \quad 3 \times 3 \times 256$

$64 \times 64 \times 3 \times 3 \times 256 \times 256 = \text{huge}$

1x1

$64 \times 64 \times 256 \quad \oplus \quad 1 \times 1 \times 256$

$64 \times 64 \times 256 \times 1 \times 1 \times 256 \Rightarrow$ 4


192
64
64
192
1X1
64
1
64
P_{11}
P_{12}
P_{21}
C_1
C_2
$(P_{11} \times C_1)$
$(P_{21} \times C_2)$ . . . . $(P_{192} \times C_{192})$
(+ + + + +)

